HCI_Track_Student_Tasks

🎯 Student Tasks: HCI Track for the AI School Handbook Assistant

Goal:
Learn how to understand real user needs, choose appropriate needfinding methods, collect useful evidence, turn findings into requirements, design a simple prototype, evaluate it with users, and improve it through iteration.

📝 Weekly Report Requirement

Each week, you must submit a short written report documenting your progress, decisions, and reflections.

📌 **Format requirement**

- Use the **JDF (Joyner Document Format)** template in **Overleaf**
- If you do not have an Overleaf account, register for free at https://www.overleaf.com
- JDF template: https://www.overleaf.com/latex/templates/jdf-for-cs-6750/vkqbhfrgxtgs

Your report should include:

- what you worked on this week
- key decisions you made and why
- any challenges or open questions
- what you plan to do next

🗂️ Suggested Workflow

Week 1 — Understand and Plan

- Watch the Needfinding videos

- Choose your primary user group
- Investigate 2–3 existing solutions
- Decide which needfinding methods to use
- Define your research goals
- Submit Week 1 report (JDF format via Overleaf)

Week 2 — Design, Collect, and Analyze

- Watch the data collection tips and need representation videos (5 Tips: Interviews, 5 Tips: Surveys, Writing Good Survey Questions, Representing the Need, Defining the Requirement)
- Draft your survey or interview guide
- Pilot and revise your questions
- Conduct interviews, surveys, or focus groups
- Analyze findings
- Identify user needs, pain points, and trust concerns
- Turn findings into initial requirements
- Submit Week 2 report (JDF format via Overleaf)

Week 3 — Prototype and Evaluate

- Watch the Design Alternatives, Prototyping, and selected Design Principles videos
- Create a simple prototype (mockup, not code)
- Prepare evaluation tasks
- Test the prototype with users using think-aloud
- Record usability and trust-related issues
- Submit Week 3 report (JDF format via Overleaf)

Week 4 — Iterate and Reflect

- Review the most important problems
- Revise the design

- Test the updated version again, if possible
- Write a short reflection on what improved
- Submit Week 4 report (JDF format via Overleaf)

1️⃣ Needfinding

🎥 What to watch

Focus on these parts of **CS6750 HCI – 3.3 Needfinding and Requirements Gathering**:

- 3.3.1 Introduction to Needfinding
- 3.3.3 The Problem Space
- 3.3.4 User Types
- 3.3.6 Naturalistic Observation
- 3.3.8 Participant Observation
- 3.3.11 Apprenticeship and Ethnography
- 3.3.12 Interviews and Focus Group
- 3.3.15 Think-aloud
- 3.3.16 Surveys
- 3.3.20 Other Data Gathering Methods

🧠 What to learn

After watching these sections, you should understand:

A. How to explore the problem space

- who the target users are
- what handbook-related questions they commonly have
- how they currently look for answers (e.g. school website, asking a teacher, searching online)
- what difficulties or frustrations they experience
- what makes them trust or distrust an AI-generated answer

B. Methods for needfinding

- what methods can be used for needfinding
- when to use interviews, surveys, or focus groups
- what kind of information each method can help you gather
- how to choose participants who match your target users

C. How to turn findings into requirements

- how to identify patterns in user responses
- how to represent user needs clearly
- how to summarize pain points and trust concerns
- how to translate findings into system requirements

✅ Task / What to do

Step 1: Choose your primary users

Recruit a small group of target users, such as:

- students
- parents
- teachers

Choose **one** primary user group to focus on for your research.
You may mention other stakeholders, but your work should mainly focus on one group.

📌 **Suggested group size**

- For interviews: 3–5 participants
- For surveys: 10–20 respondents
- For focus groups: 4–6 participants per session

Step 2: Investigate the current landscape

Before designing your research, briefly explore how your target users currently find handbook information. For example:

- look at the school website or student portal
- review any existing FAQ pages or chatbot tools

- note 2–3 existing solutions and their strengths and weaknesses

This will help you ask better questions and identify gaps.

Step 3: Choose your needfinding methods

Decide which methods you will use, such as:

- interviews
- surveys
- focus groups

Choose your methods based on what you want to learn. For example:

- use **interviews** to understand detailed experiences, frustrations, and trust concerns
- use **surveys** to gather broader patterns across more users
- use **focus groups** to compare different views and identify shared concerns

Step 4: Define your research goals

Before writing any questions, clearly state what you want to learn. For example:

- what handbook topics are most confusing or hard to find?
- what does “trust” in an AI-generated answer mean to users?
- when do users give up looking for an answer?

These goals will guide the design of your survey or interview in **Task 2**.

⚠️ **Note on ethics**

When involving real participants, always explain the purpose of the study, get their verbal or written consent, keep responses anonymous, and make participation voluntary. You do not need formal ethics board approval for this course project, but you should treat participants respectfully.

📦 Output

Create a short summary that includes:

- your target user group and why you chose them
- a brief review of 2–3 existing solutions and their limitations
- the needfinding methods you plan to use and why

- your main research goals

2️⃣ Tips for Data Collection and Need Representation

🎥 What to watch

Focus on these parts of **CS6750 HCI – 3.3 Needfinding and Requirements Gathering**:

- 3.3.13 5 Tips: Interviews
- 3.3.17 5 Tips: Surveys
- 3.3.18 Writing Good Survey Questions
- 3.3.25 Representing the Need
- 3.3.3 Defining the Requirement

🧠 What to learn

After watching these sections, you should understand:

- how to write clear survey or interview questions
- how to avoid biased, vague, leading, or double-barreled questions
- how to design questions that match your research goals
- how to revise weak questions if the first version is unclear or ineffective
- how to represent user needs in a structured and communicable form
- how to translate needs into clear, actionable requirements

✅ Task / What to do

Step 1: Design your instrument

Design a short survey or interview guide for your target users. Decide what information you want to collect and write questions that help you understand:

- user needs
- current search behavior

- common difficulties
- trust in AI answers

Step 2: Pilot and revise

- Review your questions: check for vague, leading, or double-barreled wording
- Run a small pilot test with 1–2 people
- Revise the wording based on feedback before full deployment

Step 3: Collect data

Use your finalized instrument to gather data from your target users.

Step 4: Analyze findings and define requirements

Review your findings and identify:

- repeated question types
- common frustrations
- trust-related concerns
- patterns across users

Use your results to define:

- **functional requirements** — what the system should do
- **trust and usability requirements** — what it must do well for users to trust and use it confidently

📦 Output

Produce:

- a short survey draft or interview guide
- a brief note explaining what the questions are meant to find out
- a short note describing any revisions you made and why
- a summary of findings, including:
  - common handbook question types
  - major pain points
  - trust requirements

- initial system requirements (both functional and trust-related)

3️⃣ Prototyping

🎥 What to watch

Focus on:

- **CS6750 HCI – Lesson 3.4: Design Alternatives** (for generating different interface ideas)
- **CS6750 HCI – Lesson 3.5: Prototyping** (for building and representing your design)
- **CS6750 HCI – 2.5 Design Principles and Heuristics** (selected principles for guiding your design):
  - 2.5.5 Simplicity
  - 2.5.6 Affordance
  - 2.5.10 Perceptibility
  - 2.5.11 Consistency
  - 2.5.17 Constraints

🧠 What to learn

After watching these sections, you should understand:

- how to generate multiple design ideas before committing to one
- how to turn user needs and requirements into interface ideas
- how to create a simple prototype before building the full system
- how to represent the main interaction flow clearly
- how to design for usability, clarity, and trust
- how key design principles apply to your prototype:
  - **Simplicity** — keep the interface focused and avoid unnecessary complexity
  - **Affordance** — make interactive elements (e.g. buttons, input fields) look and behave as users expect

- **Perceptibility** — make important information (e.g. sources, confidence cues) easy to notice
- **Consistency** — use consistent layout, wording, and interaction patterns throughout
- **Constraints** — guide users toward valid actions and prevent errors

✅ Task / What to do

Create a simple prototype based on your needfinding results and requirements. Design a basic interface that allows users to:

- enter a question
- view an answer
- view information that supports trust in the answer
- understand what happens when no good answer is found

Keep the design simple and focused on usability and trust.

Trust-supporting information may include:

- supporting handbook sources
- relevant excerpts
- uncertainty or confidence cues
- update information
- suggestions for what to do next if no answer is found

📌 **Prototype fidelity guidance**

At this stage, you do **not** need to write any code. Focus on representing the interaction flow and layout, not on building a working system.

- **Paper sketches** are fine for early exploration
- **Digital mockups** (e.g. Figma, Balsamiq, or similar tools) are recommended for sharing and testing
- The goal is to make your design testable, not polished

📦 Output

Produce:

- a low-fidelity prototype or interactive mockup (not a coded application)
- a short explanation of the main interaction flow
- a brief note explaining how the design reflects your user requirements

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4️⃣ Evaluation

🎥 What to watch

Focus on:

- **CS6750 HCI – Lesson 3.6: Evaluation**

🧠 What to learn

After watching this section, you should understand:

- how to test a prototype with users
- how to observe whether users can complete realistic tasks
- how to identify confusion, trust issues, and usability problems
- how to collect evidence for improvement
- how to use the **think-aloud protocol** to capture user reasoning during tasks

✅ Task / What to do

Step 1: Define your evaluation goals

Decide what you want to learn from the test, such as:

- whether users can find answers to common handbook questions
- whether they understand where the answer came from
- whether they know what to do when the system is uncertain
- whether they trust the result enough to use it

Step 2: Choose your evaluation method

Use **task-based usability testing** as your primary method. During testing:

- ask participants to **think aloud** — verbalize what they are thinking, expecting, and noticing as they use the prototype

- this helps you understand not just what users do, but why

Optionally, you may also conduct a brief **heuristic evaluation** (reviewing the design against known usability principles) as a supplement.

Step 3: Prepare realistic tasks

Create a few tasks based on real handbook use cases, such as:

- finding the attendance policy
- checking what to do about a missed assignment
- looking up a dress code rule
- deciding whether an answer is trustworthy

Step 4: Test the prototype with users

Recruit **3–5 users** from your target group and observe how they interact with the design.

Record:

- whether they can complete the task
- how long it takes
- where they get confused
- whether they understand the trust cues
- whether they trust the answer
- what they say during think-aloud

📌 **How to record observations**

- Take notes during each session (a simple table with columns for task, success, time, and comments works well)
- Screen recording is helpful but not required
- After each session, write a short summary while details are fresh

Step 5: Identify issues

Summarize the main usability and trust-related problems that appeared during testing.

📦 Output

Produce:

- a short evaluation summary
- a list of usability problems
- a list of trust-related issues
- a list of improvements that should be prioritized

5️⃣ Iteration

🎥 What to review

No new video is required for this task.
If needed, revisit the sections on identifying usability issues and prioritizing improvements in **CS6750 HCI – Lesson 3.6: Evaluation**.

Use the overall HCI process:

Needfinding → Requirements → Prototyping → Evaluation → Iteration

🧠 What to learn

From this stage, you should understand:

- how to improve a design based on evidence
- how to prioritize the most important problems
- how to revise a prototype step by step
- how to use evaluation results to guide the next version

✅ Task / What to do

- review the problems found during evaluation
- choose a few **high-priority issues** to fix
- revise the prototype
- test the updated version again with a small number of users (2–3), if possible
- reflect on whether the changes improved the experience

Focus first on issues that strongly affect:

- task completion
- clarity
- trust
- user confidence

📦 Output

Produce:

- a revised prototype
- a short change log describing:
  - what was changed
  - why it was changed
  - what evidence motivated the change
  - whether the revision improved the experience (based on re-testing or your own reflection)

📋 Evaluation Criteria

Each task will be assessed on the following dimensions:

Task	What makes it strong
1. Needfinding	Based on a clear user group; includes landscape review; research goals are specific and relevant
2. Data Collection & Need Representation	Questions are clear, unbiased, and aligned with research goals; instrument design choices are justified; revisions are explained; findings are grounded in real data; needs are clearly represented; requirements cover both function and trust
3. Prototyping	Reflects the requirements; interaction flow is clear; includes trust-supporting elements; applies relevant design principles (simplicity, affordance, perceptibility, consistency, constraints); appropriate fidelity
4. Evaluation	Uses realistic tasks; involves real users; observations are specific and evidence-based; issues are clearly identified
5. Iteration	Prioritizes the right problems; changes are justified by evidence; reflection is honest and thoughtful

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✨ Final Reminder for Students

- Do not start with coding.
- Start by understanding the users, the problem space, the most suitable needfinding methods, and what makes the system trustworthy.
- Your prototype should come from evidence, not assumptions.
- Treat your participants with respect: explain the study, get consent, and keep responses anonymous.

🪄 One-line Summary

Use needfinding methods to understand real user needs, turn your findings into requirements, design a simple prototype, test it with users, and improve it based on evidence.